

HetSched: Quality-of-Mission Aware Scheduling for Autonomous Vehicle SoCs

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Abstract—Systems-on-Chips (SoCs) that power autonomous vehicles (AVs) must meet stringent performance and safety requirements prior to deployment. With increasing complexity in AV applications, the system needs to meet stringent real-time demands of multiple safety-critical applications simultaneously. A typical AV-SoC is a heterogeneous multiprocessor consisting of accelerators supported by general-purpose cores. Such heterogeneity, while needed for power-performance efficiency, complicates the art of task (process) scheduling.

In this paper, we demonstrate that hardware heterogeneity impacts the scheduler’s effectiveness and that optimizing for only the real-time aspect of applications is not sufficient in AVs. Therefore, a more holistic approach is required — one that considers global *Quality-of-Mission* (QoM) metrics, as defined in the paper. We then propose HetSched, a multi-step scheduler that leverages dynamic runtime information about the underlying heterogeneous hardware platform, along with the applications’ real-time constraints and the task traffic in the system to optimize overall mission performance. HetSched proposes two scheduling policies: MS_{stat} and MS_{dyn} and scheduling optimizations like task pruning, hybrid heterogeneous ranking and rank update. HetSched improves overall mission performance on average by $4.6\times$, $2.6\times$ and $2.6\times$ when compared against CPATH, ADS and 2lvi-EDF (state-of-the-art real-time schedulers built for heterogeneous systems), respectively, and achieves an average of 53.3% higher hardware utilization, while meeting 100% critical deadlines for real-world applications of autonomous driving and aerial vehicles. Furthermore, when used as part of an SoC design space exploration loop, in comparison to the prior schedulers, HetSched reduces the number of processing elements required by an SoC to safely complete AV’s missions by 35% on average while achieving $2.7\times$ lower energy-mission time product.

I. INTRODUCTION

With the growing prominence of fully autonomous vehicles (ground, aerial, surface and underwater), major investments are being made into developing applications to make these vehicles efficient and safe. In order to ensure functionally correct and safe operation, the complexity of state-of-the-art full-stack hardware-software platforms for autonomous vehicles (AVs) has progressively increased over the last decade — specifically in the form of highly-heterogeneous hardware systems driven by highly-heterogeneous software applications.

The resulting “nominal” hardware-software platform for AVs consists of domain-specific systems-on-chips (DSSoCs) with multiple acceleration engines specifically catering to ultra-efficient execution of application kernels for perception, planning, control, and communication. Examples of these platforms include NVIDIA’s DRIVE AGX [7] and Tesla’s Full Self-Driving (FSD) chip [49]. In this context, the existing work focuses mostly on: (i) optimal SoC platforms for AVs to comply with stipulated performance, efficiency and resiliency metrics, and (ii) the development of AV applications that can meet the increasing functionality and safety requirements for autonomy. Little attention has been paid to the aspect of process scheduling for AV applications on heterogeneous DSSoCs. The *de facto* approach relies on schedulers developed for:

- 1) Heterogeneous systems (traditional distributed systems and not SoCs, therefore low variation in execution time across processing elements) [51], [21].
- 2) Real-time constrained applications [3], [4].
- 3) Real-time constrained applications based on the ones listed in 1) above [58], [56], [57].

However, none of these schedulers use dynamic runtime information from the system to efficiently schedule real-time applications on heterogeneous SoCs. Moreover, these schedulers operate in a greedy manner, trying to meet the real-time requirements of individual processes or applications without any consideration of a *global* objective function, defined as the Quality-of-Mission (QoM) metric for AV applications. We define the QoM as a composite metric encompassing both the mission performance, the fraction of mission completed *safely*, i.e. while meeting the real-time and safety constraints and the energy consumed.

To demonstrate the value of considering the QoM, we examine the following two schedulers:

- 1) **Quality-of-Mission Agnostic Approach.** The tasks of an AV application, represented as directed-acyclic graphs (DAG), are statically scheduled using the earliest-finish-